

- 1 (a) Find the range of values of x for which the expression $3 - 2x^2$ is negative. [2]

$$3 - 2x^2 < 0$$

$$2x^2 - 3 > 0$$

$$x < -\sqrt{\frac{3}{2}} \quad \text{or} \quad x > \sqrt{\frac{3}{2}} \quad \text{OR}$$

$$x < -\frac{\sqrt{6}}{2} \quad \text{or} \quad x > \frac{\sqrt{6}}{2}$$

M1: finding roots or factorising

A1

- (b) Find the set of values of the constant k for which the curve $y = x^2$ lies entirely above the line $y = k(x + 1)$. [3]

$$x^2 > k(x + 1)$$

$$x^2 - kx - k > 0$$

For the quadratic expression to be always positive,

$$b^2 - 4ac < 0$$

$$k^2 + 4k < 0$$

M1

$$k(k + 4) < 0$$

M1

$$-4 < k < 0$$

A1

- 2 (a) Find the range of values of k such that the line $x + y = 3$ intersects the curve $x^2 - 2x + 2y^2 = k$. [4]

$$x + y = 3$$

$$y = 3 - x \text{-----}(1)$$

$$x^2 - 2x + 2(3 - x)^2 = k$$

$$x^2 - 2x + 2(9 - 6x + x^2) = k$$

$$x^2 - 2x + 18 - 12x + 2x^2 - k = 0$$

$$3x^2 - 14x + 18 - k = 0$$

Since the line and curve intersects,

$$b^2 - 4ac \geq 0$$

$$(-14)^2 - 4(3)(18 - k) \geq 0$$

$$196 - 216 + 12k \geq 0$$

$$-20 + 12k \geq 0$$

$$k \geq \frac{5}{3}$$

M1 manipulation to get quadratic equation in 1 unknown

M1

M1

A1

- (b) State a possible value of k if there is no intersection between the line and the curve. [1]

Any value that is $< \frac{5}{3}$.

B1

- 3 A polynomial, P , is $x^{2n} - (k+1)x^2 + k$ where n and k are positive integers.

(a) Explain why $x-1$ is a factor of P for all values of k . [2]

$$\text{let } f(x) = x^{2n} - (k+1)x^2 + k$$

$$f(1) = 1 - k - 1 + k$$

$$= 0$$

$\therefore$ since remainder $= 0$, $(x-1)$ is a factor.

M1

A1: must mention remainder $= 0$,
or by factor theorem

(b) Given that $k = 4$, find the value of n for which $x-2$ is a factor of P .
Hence factorise P completely. [4]

$$f(x) = x^{2n} - 5x^2 + 4$$

$$f(2) = 2^{2n} - 16$$

Since $x-2$ is a factor,

$$2^{2n} - 16 = 0$$

$$n = 2$$

$$f(x) = x^4 - 5x^2 + 4$$

$$= (x-1)(x-2)(x^2 + 3x + 2)$$

$$= (x-1)(x-2)(x+1)(x+2)$$

M1

A1

M1, must write factors $(x-1)(x-2)$ first since
it's a hence question

A1

- 4 A projectile was launched from a catapult to hit a defence structure on a fort. The height, h metres, of the projectile above ground is given by the equation $h = -2x^2 + 3x + 1.5$, where x metres is the horizontal distance from the catapult.

(i) By expressing the function in the form $h = a(x - m)^2 + n$, where a , m and n are constants, explain whether the projectile can reach a height of 3 metres. [2]

$$\begin{aligned} h &= -2x^2 + 3x + 1.5 \\ &= -2(x^2 - 1.5x - 0.75) \\ &= -2[(x - 0.75)^2 - 0.75^2 - 0.75] \\ &= -2(x - 0.75)^2 - 2.625 \end{aligned}$$

maximum point is (0.75, 2.625)

Therefore the projectile cannot reach a height of 3m since the maximum height is 2.625m.

M1 for completing the square

A1 for comparing 2.625 and 3

- (ii) Given that the defence structure is 1.4 metres horizontally from the catapult and 0.8 metres above the ground, justify if the projectile will hit the structure. [2]

Subs (1.4, 0.8) into $h = -2x^2 + 3x + 1.5$

$$\begin{aligned} h &= -2(1.4)^2 + 3(1.4) + 1.5 \\ &= 1.78 \\ &\neq 0.8 \end{aligned}$$

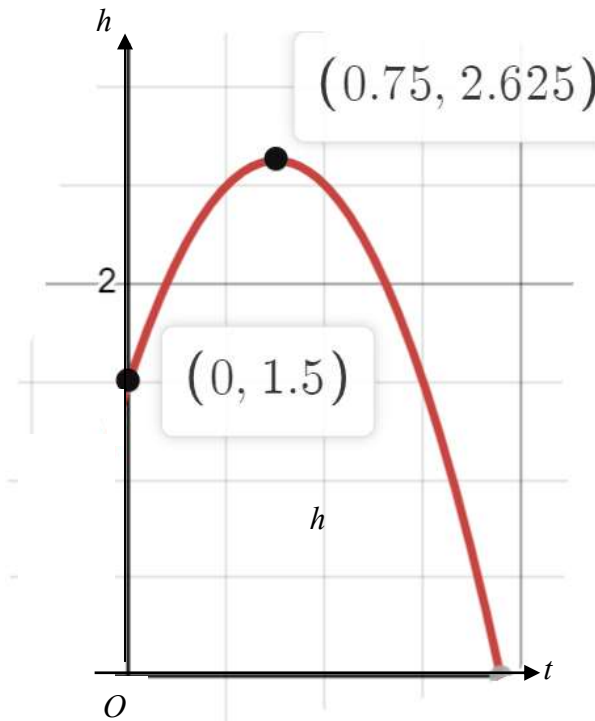
M1 for determining if the point lies on the equation

Since the point (1.4, 0.8) does not lie on the curve $h = -2x^2 + 3x + 1.5$, therefore the projectile will not hit the structure.

A1

(iii) Sketch the curve of $h = -2x^2 + 3x + 1.5$.

[2]



B! correct shape with turning point

B1 correct y-intercept

[Turn over

- 5 (a) Differentiate $\ln(\sin x)$ with respect to x .

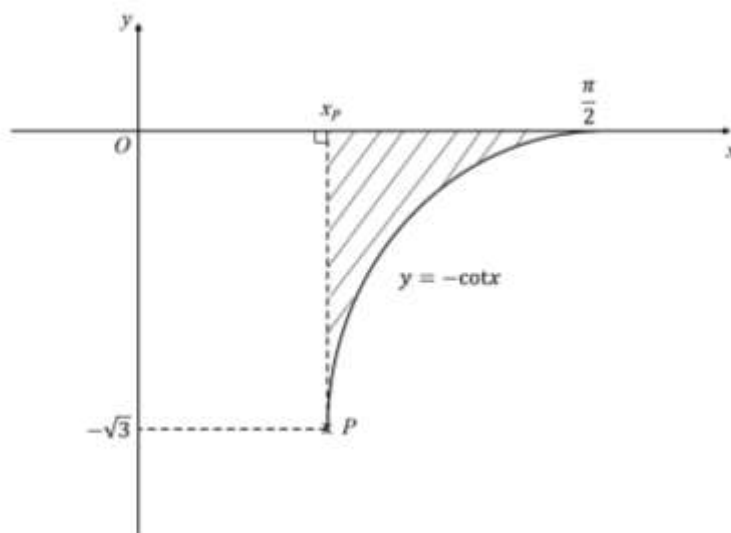
[2]

$$\begin{aligned}\frac{d}{dx} \ln(\sin x) \\ &= \frac{1}{\sin x} \cos x \\ &= \cot x\end{aligned}$$

M1

A1

(b)



The diagram shows part of the curve $y = -\cot x$, cutting the x -axis at $\left(\frac{\pi}{2}, 0\right)$.

The line $y = -\sqrt{3}$ intersects the curve at P .

- (i) State the value of x_P , the x -coordinate of P .

[1]

$$\sqrt{3} = \frac{1}{\tan x}$$

$$\tan x = \frac{1}{\sqrt{3}}$$

$$x = \frac{\pi}{6}$$

B1

- (ii) Explain why the expression $\int_{x_p}^{\frac{\pi}{2}} -\cot x \, dx$ does not give the area of the shaded region. [1]

The shaded area is below the x -axis. If we $\int_{Q_x}^{\frac{\pi}{2}} -\cot x \, dx$, we will get a **negative value** for the area. Thus $\int_{Q_x}^{\frac{\pi}{2}} -\cot x \, dx$ does not give area of the shaded region.

B1

- (iii) Find the exact area of the shaded region. [3]

$$y = -\frac{1}{\tan x}$$

when $y = -\sqrt{3}$

$$-\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cot x \, dx$$

M1

$$= [\ln(\sin x)]_{\frac{\pi}{6}}^{\frac{\pi}{2}}$$

M1

$$= \ln 1 - \ln \frac{1}{2}$$

$$= \ln 2 \text{ units}^2 \text{ or } -\ln \frac{1}{2} \text{ units}^2$$

A1

- 6 (a) Without using a calculator, show that $\cos\left(\frac{7\pi}{12}\right) = \frac{1}{4}(\sqrt{2} - \sqrt{6})$. [3]

$$\begin{aligned}
 \cos\left(\frac{7\pi}{12}\right) &= \cos\left(\frac{\pi}{4} + \frac{\pi}{3}\right) && \text{M1: identifying special angles} \\
 &= \cos\left(\frac{\pi}{4}\right)\cos\left(\frac{\pi}{3}\right) - \sin\left(\frac{\pi}{4}\right)\sin\left(\frac{\pi}{3}\right) && \text{M1: correct application of formula} \\
 &= \frac{\sqrt{2}}{2}\left(\frac{1}{2}\right) - \frac{\sqrt{2}}{2}\left(\frac{\sqrt{3}}{2}\right) \\
 &= \frac{1}{4}(\sqrt{2} - \sqrt{6}) && \text{M1: recognising exact values and reach result given}
 \end{aligned}$$

- (b) Evaluate $\int_0^{\frac{\pi}{12}} 3\cos^2 x - \sin^2 x \, dx$ exactly. [4]

$$\begin{aligned}
 \int_0^{\frac{\pi}{12}} 3\cos^2 x - \sin^2 x \, dx &= \int_0^{\frac{\pi}{12}} \frac{3}{2}(2\cos^2 x - 1 + 1) + \frac{1}{2}(1 - 2\sin^2 x - 1) \, dx \\
 &= \int_0^{\frac{\pi}{12}} \frac{3}{2}\cos 2x + \frac{1}{2}\cos 2x + 1 \, dx && \text{M1: correct application of double angle formula} \\
 &= \int_0^{\frac{\pi}{12}} 2\cos 2x + 1 \, dx && \text{M1} \\
 &= [\sin 2x + x]_0^{\frac{\pi}{12}} \\
 &= \sin \frac{\pi}{6} + \frac{\pi}{12} && \text{M1} \\
 &= \frac{6 + \pi}{12} && \text{A1}
 \end{aligned}$$

- 7 (a) (i) Factorise $x^6 - 64$ completely. [2]

$$\begin{aligned} x^6 - 64 &= (x^2)^3 - 4^3 \\ &= (x^2 - 4)(x^4 + 4x^2 + 16) \\ &= (x - 2)(x + 2)(x^4 + 4x^2 + 16) \end{aligned}$$

M1: either cubic factorisation or
difference of squares factorisation

A1

OR

$$\begin{aligned} &(x^3)^2 - (2^3)^2 \\ &= (x^3 - 8)(x^3 + 8) \\ &= (x^3 - 2^3)(x^3 + 2^3) \\ &= (x - 2)(x^2 + 2x + 4)(x + 2)(x^2 - 2x + 4) \\ &= (x - 2)(x + 2)(x^2 + 2x + 4)(x^2 - 2x + 4) \end{aligned}$$

M1

A1

- (ii) Hence solve $x^6 - 64 = (x^2 + 4)^2 - (2x)^2$. [3]

$$\begin{aligned} x^6 - 64 &= (x^2 + 4)^2 - (2x)^2 \\ (x + 2)(x - 2)(x^4 + 4x^2 + 16) &= x^4 + 8x^2 + 16 - 4x^2 \\ (x + 2)(x - 2)(x^4 + 4x^2 + 16) &= x^4 + 4x^2 + 16 \\ (x + 2)(x - 2) &= 1 \\ x^2 - 5 &= 0 \\ x &= \pm\sqrt{5} \end{aligned}$$

M1 for expanding
the RHS of the
equation

M1

A1

[Turn over

- (b) Find the values of the integers a and b for which $\frac{a + \sqrt{b}}{2}$ is the solution of the equation

$$2x\sqrt{3} + x\sqrt{125} = x\sqrt{45} + \sqrt{12}. \quad [4]$$

$$2x\sqrt{3} + x\sqrt{125} = x\sqrt{45} + \sqrt{12}$$

$$2x\sqrt{3} + 5x\sqrt{5} = 3x\sqrt{5} + 2\sqrt{3}$$

$$2x\sqrt{3} + 5x\sqrt{5} - 3x\sqrt{5} = 2\sqrt{3}$$

$$x(2\sqrt{3} + 2\sqrt{5}) = 2\sqrt{3}$$

$$x = \frac{2\sqrt{3}}{(2\sqrt{3} + 2\sqrt{5})} \times \frac{(2\sqrt{3} - 2\sqrt{5})}{(2\sqrt{3} - 2\sqrt{5})}$$

$$= \frac{12 - 4\sqrt{15}}{12 - 20}$$

$$= \frac{12 - 4\sqrt{15}}{-8}$$

$$= \frac{-3 + \sqrt{15}}{2}$$

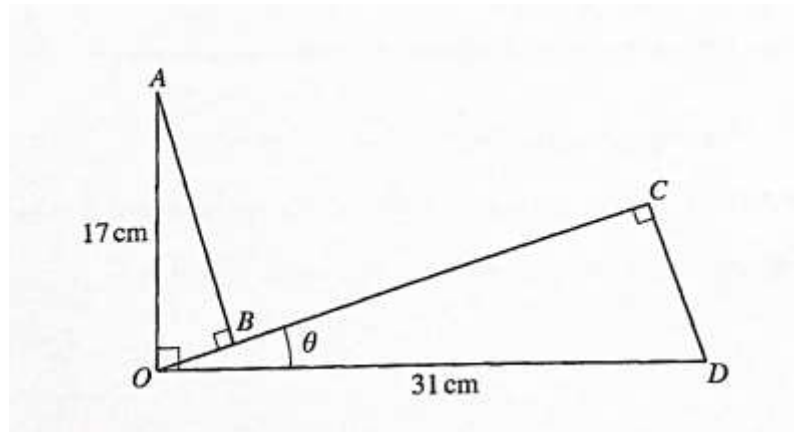
$$a = -3, b = 15$$

M1 , isolating x terms and
simplifying the surds

M1 for rationalising

A1 , A1

8



The diagram shows three fixed points O , A and D such that $OA = 17$ cm, $OD = 31$ cm and angle $AOD = 90^\circ$.

The lines AB and DC are perpendicular to the line OC which makes an angle θ with the line OD .

The angle θ can vary in such a way that the point B lies between the points O and C .

- (i) Show that $AB + BC + CD = (48 \cos \theta + 14 \sin \theta)$ cm. [3]

$$\sin \theta = \frac{CD}{31}$$

$$CD = 31 \sin \theta$$

$$\sin \theta = \frac{OB}{17}$$

$$OB = 17 \sin \theta$$

$$\cos \theta = \frac{AB}{17}$$

$$AB = 17 \cos \theta$$

$$\cos \theta = \frac{OC}{31}$$

$$OC = 31 \cos \theta$$

$$AB + BC + CD$$

$$= 17 \cos \theta + 31 \cos \theta - 17 \sin \theta + 31 \sin \theta$$

$$= (48 \cos \theta + 14 \sin \theta) \text{ cm}$$

M2 for any 2 correct

A1

[Turn over

(ii) Find the values of θ for which $AB + BC + CD = 49$ cm.

[6]

$$48 \cos \theta + 14 \sin \theta = 49$$

$$R \cos(\theta - \alpha) = 49$$

M1

$$R = \sqrt{48^2 + 14^2}$$

$$= 50$$

M1

$$\alpha = \tan^{-1}\left(\frac{14}{48}\right)$$

$$= 16.26^\circ$$

M1

$$50 \cos(\theta - 16.26^\circ) = 49$$

$$\cos(\theta - 16.26^\circ) = \frac{49}{50}$$

M1

$$\text{Reference angle} = 11.48^\circ$$

$$\theta = 27.7^\circ, 4.8^\circ$$

A1, A1

(iii) Find the maximum value of $AB + BC + CD$ and the corresponding value of θ .

[2]

$$\max 50 \cos(\theta - 16.26^\circ)$$

$$= 50$$

B1

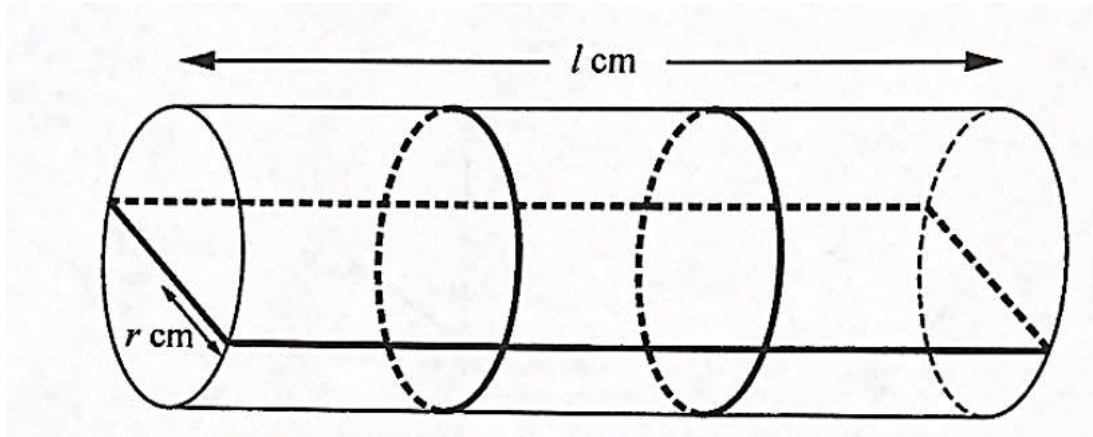
$$\text{occurs when } \cos(\theta - 16.26^\circ) = 1$$

$$\theta - 16.26^\circ = 0$$

$$\theta = 16.3^\circ \text{ (1 d.p.)}$$

B1

- 9 The diagram shows a roll of material in the shape of a cylinder of radius r cm and length l cm.
 The roll is held together by three pieces of adhesive tape whose width and thickness may be ignored.
 One piece of tape is in the shape of a rectangle, the other two pieces are in the shape of circles.
 The total length of tape is 600 cm.



- (i) Show that the volume, V cm³, of the cylinder is given by
 $V = \pi r^2 (300 - 2r - 2\pi r)$. [3]

$$\begin{aligned}
 2(2r) + 2(l) + 2(2\pi r) &= 600 \\
 2r + l + 2\pi r &= 300 \\
 l &= 300 - 2r - 2\pi r \\
 V &= \pi r^2 l \\
 &= \pi r^2 (300 - 2r - 2\pi r) \text{ (shown)}
 \end{aligned}$$

M1

M1

A1

- (ii) Given that r can vary, show that V has a stationary value when $r = \frac{k}{1+\pi}$, where k is a constant to be found, and find the corresponding value of l . [5]

$$\begin{aligned}\frac{dV}{dr} &= 600\pi r - 6\pi r^2 - 6\pi^2 r^2 \\ &= 6\pi r(100 - r - \pi r) \quad \text{M1}\end{aligned}$$

For stationary value, $\frac{dV}{dr} = 0$.

$$6\pi r(100 - r - \pi r) = 0 \quad \text{M1}$$

$r = 0$ rejected because $r > 0$ or

$$100 - r - \pi r = 0$$

$$(1 + \pi)r = 100$$

$$r = \frac{100}{1 + \pi} \quad \text{A1}$$

$\therefore V$ has a stationary value when $r = \frac{100}{1 + \pi}$, where $k = 100$. (shown)

$$\text{When } r = \frac{100}{1 + \pi},$$

$$l = 300 - 2\left(\frac{100}{1 + \pi}\right) - 2\pi\left(\frac{100}{1 + \pi}\right) \quad \text{M1}$$

$$= 300 - 2\left(\frac{100}{1 + \pi}\right)(1 + \pi)$$

$$= 300 - 200$$

$$= 100 \quad \text{A1}$$

- (iii) Determine if the volume is a minimum or maximum. [3]

$$\frac{dV}{dr} = 6\pi r(100 - r - \pi r)$$

$$\frac{d^2V}{dr^2} = 6\pi r(-1 - \pi) + (100 - r - \pi r)(6\pi)$$

$$= 6\pi r(-1 - \pi) + (100 - r - \pi r)(6\pi)$$

$$= 6\pi(-r - \pi r + 100 - r - \pi r)$$

$$= 6\pi(-2r - 2\pi r + 100)$$

M1

$$\text{when } r = \frac{100}{1 + \pi},$$

$$\frac{d^2V}{dr^2} = 6\pi\left(-\frac{200}{1 + \pi} - \frac{200\pi}{1 + \pi} + 100\right)$$

$$= -600\pi$$

M1

Since $\frac{d^2V}{dr^2} < 0$, V is a maximum.

A1

- 10** A particle travelling in a straight line passes through a fixed point O with a speed of 8 m/s.

The acceleration, $a \text{ m/s}^2$, of the particle t s after passing through O , is given by $a = -e^{-0.1t}$.
The particle comes to instantaneous rest at the point P .

- (i) Show that the particle reaches P when $t = 10 \ln 5$. [5]

$$\begin{aligned}
 a &= -e^{-0.1t} \\
 v &= \int a = -e^{-0.1t} \, dt \\
 &= \frac{-e^{-0.1t}}{-0.1} + c \\
 \text{When } t = 0, v &= 8 \\
 8 &= \frac{-1}{-0.1} + c \\
 8 &= 10 + c \\
 c &= -2 \\
 v &= 10e^{-0.1t} - 2 \\
 \text{when } v &= 0 \\
 10e^{-0.1t} - 2 &= 0 \\
 e^{-0.1t} &= \frac{1}{5} \\
 -0.1t &= -\ln 5 \\
 t &= 10 \ln 5 \text{ (shown)}
 \end{aligned}$$

M1

M1

M1

M1

A1

- (ii) Calculate the distance OP . [3]

$$\begin{aligned}
 s &= \int_0^{10 \ln 5} 10e^{-0.1t} - 2 \, dt \\
 &= \left[\frac{10e^{-0.1t}}{-0.1} - 2t \right]_0^{10 \ln 5} \\
 &= -100e^{-0.1(10 \ln 5)} - 20 \ln 5 + 100 \\
 &= -100e^{-\ln 5} - 20 \ln 5 + 100 \\
 &= -20 - 20 \ln 5 + 100 \\
 &= 47.8 \text{ m}
 \end{aligned}$$

M1

M1

A1

- (iii) Explain why the particle is again at O at some instant during the fiftieth second after first passing through O . [3]

when $t = 49$

$$s = -100e^{-0.1(49)} - 2(49) + 100$$

$$= 1.255m \quad \text{M1}$$

when $t = 50$

$$s = -100e^{-0.1(50)} - 2(50) + 100$$

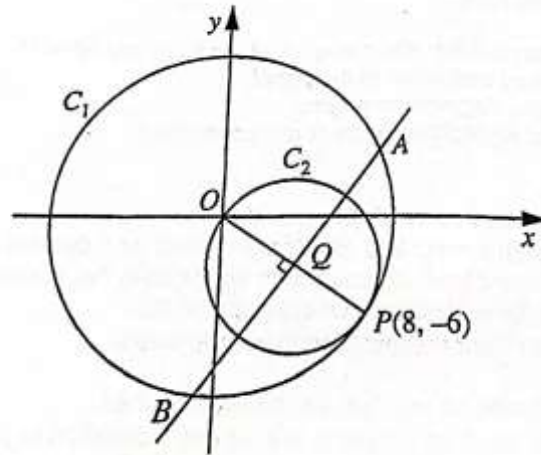
$$= -0.674m \quad \text{M1}$$

Since displacement of the particle is positive when $t = 49$ and negative when $t = 50$, this shows that the particle must have passed through O at some point in the fiftieth second.

Thus the particle is again at O at some instant during the fiftieth second after passing through O . A1

|

11



The diagram shows two circles C_1 and C_2 .

Circle C_1 has its centre at the origin O .

Circle C_2 passes through O and has its centre at Q .

The point $P(8, -6)$ lies on both circles and OP is a diameter of C_2 .

(i) Find the equation of C_1 .

[2]

$$|OP| = \sqrt{(8-0)^2 + (-6-0)^2}$$

$$= 10$$

$$\text{Equation of } C_1 : x^2 + y^2 = 100$$

M1

A1

(ii) Explain why the equation of C_2 is $x^2 + y^2 - 8x + 6y = 0$.

[3]

$$x^2 + y^2 - 8x + 6y = 0$$

$$x^2 - 8x + y^2 + 6y = 0$$

$$(x-4)^2 - 16 + (y+3)^2 - 9 = 0$$

$$(x-4)^2 + (y+3)^2 = 5^2$$

Centre = $(4, -3)$ because it is the mid point of OP

Radius is 5 units because it is $\frac{1}{2}|OP|$

M1

A1

A1

[Turn over]

(iii) The line through Q perpendicular to OP meets the circle C_1 at the point A and

Show that the x -coordinates of A and B are $a + b\sqrt{3}$ and $a - b\sqrt{3}$ respectively, where a and b are integers to be found. [7]

$$\begin{aligned}\text{gradient } OP &= -\frac{6}{8} \\ &= -\frac{3}{4}\end{aligned}$$

$$\text{gradient } AB = \frac{4}{3}$$

$$\frac{4}{3}(x-4) = y+3$$

$$4x-16 = 3y+9$$

$$y = \frac{4}{3}x - \frac{25}{3} \dots\dots\dots(1)$$

M1

$$x^2 + y^2 = 100$$

$$y^2 = 100 - x^2 \dots\dots\dots(2)$$

$$(1)^2 \quad y^2 = \frac{16}{9}x^2 - \frac{200}{9}x + \frac{625}{9} \dots\dots\dots(3)$$

$$(2) = (3)$$

$$\frac{16}{9}x^2 - \frac{200}{9}x + \frac{625}{9} = 100 - x^2$$

$$\frac{25}{9}x^2 - \frac{200}{9}x - \frac{275}{9} = 0$$

$$x^2 - 8x - 11 = 0$$

$$x = \frac{8 \pm \sqrt{64 + 44}}{2}$$

$$= \frac{8 \pm \sqrt{108}}{2}$$

$$= \frac{8 \pm 6\sqrt{3}}{2}$$

$$= 4 \pm 3\sqrt{3}$$

$$x\text{-coordinate of } A \text{ is } 4 + 3\sqrt{3}$$

$$x\text{-coordinate of } B \text{ is } 4 - 3\sqrt{3}$$

M1 Reasonable attempt at manipulating the equations to obtain the quadratic equation

M1

M1

M1

A1, A1